

- Q.25 Simplify: $F = (A+B)(A+B+D)D$
- Q.26 Design and implement a half adder with neat diagram.
- Q.27 What do you mean by K-Map? Draw three variable K-Map.
- Q.28 Explain the diagram of 8:1 multiplexer with its truth table.
- Q.29 Explain R-S Flip-Flop.
- Q.30 Explain octal to binary encoder.
- Q.31 Differentiate between combinational and sequential Circuits.
- Q.32 Explain SISO Shift register with diagram.
- Q.33 Differentiate between RAM and ROM.
- Q.34 List the methods of Analog to Digital Converter.
- Q.35 Write the classification of semiconductor memories.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Simplify with K-Map and realize the circuit with NAND gates only.
 $Y = \sum m(0, 1, 3, 5, 7, 9, 11, 13, 15)$
- Q.37 Define Encoder. Explain its function using block diagram.
- Q.38 Explain in detail R/2R ladder Digital to Analog converter.

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4th Sem / Elect.
Subject:- Digital Electronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The binary number 10110 is equivalent to decimal number_____ -
- a) 16 b) 21
c) 20 d) 22
- Q.2 The ones complement of 111001 is _____
- a) 111000 b) 000110
c) 000011 d) 111111
- Q.3 The inputs of a NAND gate are connected together for _____ gate
- a) OR b) AND
c) NOT d) NOR
- Q.4 The logic circuit shown below is of _____ gate.



- a) NOR gate b) NAND gate
c) EX-OR gate d) EX-NOR gate

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- Q.5 Boolean algebra can be used _____
- For designing of the digital computers
 - In building logic symbols.
 - Circuit theory
 - Building algebraic functions
- Q.6 The expression for Absorption law is given by
- $A+AB=A$
 - $A+AB=B$
 - $AB+B=AB$
 - $A+B=B+A$
- Q.7 A half adder has _____ input and _____ output
- one, one
 - one, two
 - two, one
 - two, two
- Q.8 The basic building block of sequential logic circuit is _____
- OR gate
 - Flip-Flop
 - Adder
 - AND gate
- Q.9 The number of resistors required for a five bit resistor divider DAC are _____
- 10
 - 5
 - 8
 - 31
- Q.10 Which of the following memories are non volatile memories?
- ROM
 - PROM
 - Ferrite core memory
 - None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The number of digits in octal system is _____
- Q.12 2's complement of 000001 is _____
- Q.13 The output of a full adder are _____ and _____
- Q.14 $A+1=$ _____
- Q.15 SIPO stands for _____
- Q.16 Define multiplexer.
- Q.17 Basic storage elements for static RAM is _____
- Q.18 Which flip-flop is used to make all types of shift register?
- Q.19 The minimum number of flip flops required for a decade counter is _____.
- Q.20 E^2 PROM stands for _____

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain in brief the need of digitization.
- Q.22 a) Convert 110101 into gray code
b) Convert 1100 into XS-3 code (Excess-3 code)
- Q.23 Draw symbol and truth table of OR gate and AND gate.
- Q.24 Define distributive and associative law of Boolean algebra.